

iStartek Ultrasonic Fuel Sensor

UL212/UL212-E User Manual

V1.1



UL212/UL212-E

Bluetooth(BLE5.0) All-in-one Ultrasonic Fuel Level Sensor

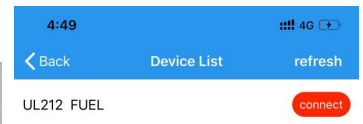
1. Indoor Test

1.1. Indoor Test Prepare

- 1) Mineral Water Bottle (flat cap)
- 2) UL212/UL212-E Kits
- 3) Android or iOS Phone
- 4) 12/24V Power Bank/Jump Starter/Power Supply

1.2. Indoor Test Steps

- 1) Power On UL212/UL212-E, RED LED flashes, and connect the UL212 Mobile APP via Bluetooth
- 2) Search Bluetooth Device ID in APP and connect "**UL212 Fuel**"



- 3) Paste 0.5~1mm thickness Silicone Gel on top of the probe



- 4) Make sure water bottle has more than 3cm height water inside, and put on the probe



- 5) Check Real Time Mobile APP data in Smart Phone.

UL212 Outdoor On-Site Installation Quick Guide

● Outdoor Installation Prepare



- 1) **Make sure fuel tank has at least 1/2 fuel inside**
- 2) Make sure there is flat zone on the bottom of tank to fix probe
- 3) UL212/UL212-E Kits
- 4) Android or iOS Phone
- 5) 12/24V Power Bank/Jump Starter or Portable Battery



or

- 6) Slot Type Screw Driver



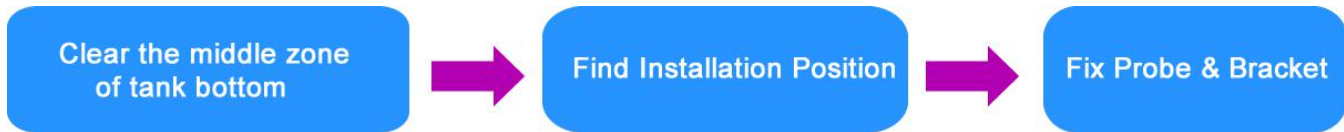
- 7) AB Epoxy Glue (**ONLY for Metal/Steel Tank**, no need for Aluminum/Fiber Plastic Tank): Suggested Ergo 9988, if cannot find, please buy AB Epoxy Glue for Metal&Plastic with 3~5 minutes Curing Time



- 8) Manual Jack/Lift (**ONLY for Metal/Steel Tank to hold probe after using AB Epoxy Glue**)

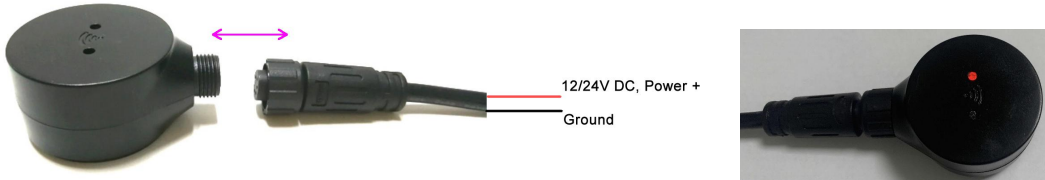


● Key Steps for Outdoor Fuel Tank Installation

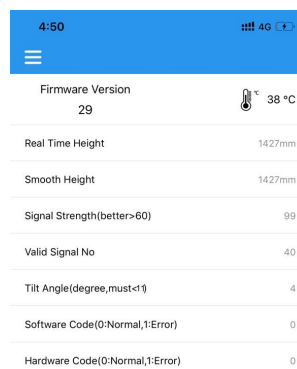
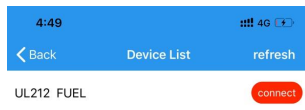


1. Power on Sensor & Connect Mobile APP “UL212 FUEL”

- 1) Power On UL212/UL212-E, RED LED flashes, and connect the UL212 Mobile APP via Bluetooth

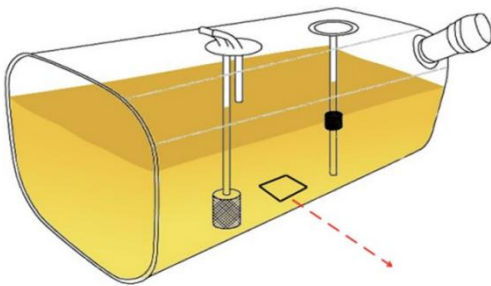


- 2) Search Bluetooth Device ID in APP and connect “UL212 FUEL”



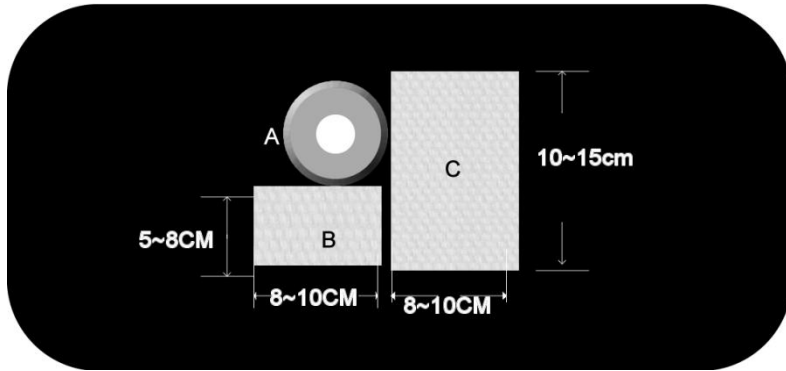
2. Clean the target zone around the middle of tank bottom

Top View from Tank Bottom



Top View from Tank Bottom





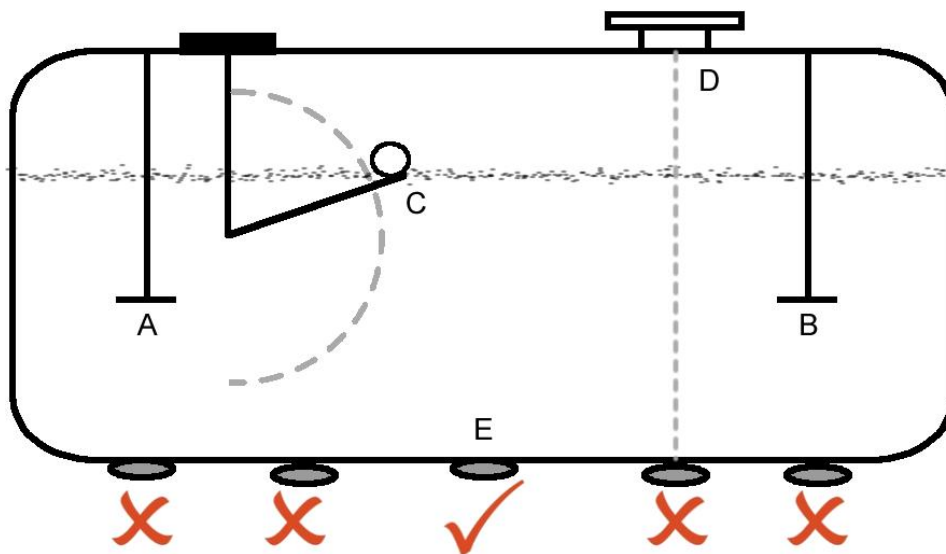
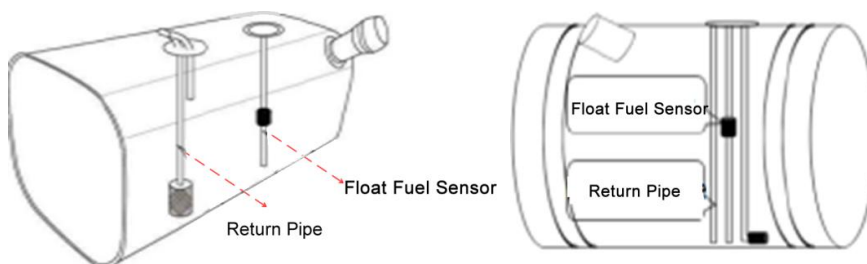
A. Tank Original Screw to drop fuel(do not touch or clean)

B. Target Zone to Clean

C. Target Zone to Clean

How to Clean?

- 1) Clean all dust, black paint, rust, oil, asphalt, etc. to see original metal material
- 2) After step 1), must use Sand Paper to polish for 2~3 minutes to make the zone as flat and smooth as possible.
- 3) Please make sure get rid of the original tank float sensor and return pipe



- A,B: Original Tank Compartment
 C: Original Tank Fuel Tube Float Range (Circle)
 D: Refueling Port
 E: Best Position for installation

3. Take notes of Vehicle Original Fuel Gauge and Measure Tank Height

- Fuel Gauge: ____ 1/4 ____ 1/2 ____ 3/4 ____ 4/4
- TANK Maximum Height: _____ mm
- Inside Diesel/Gasoline Height: _____ mm



4. Find Best Installation Position

- After clean zone B & C, use **Silicone Gel/Car grease** on top of the probe.

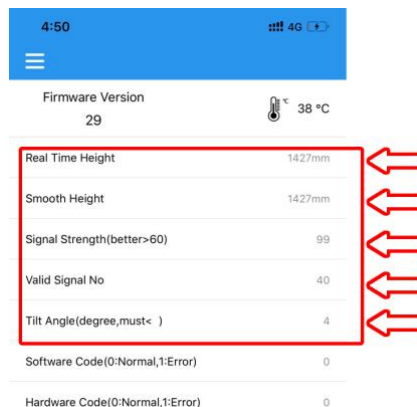


- Then hold the probe to the tank bottom and move slowly in **zone B & C**
 - Make remark with mark pen or sticker to remember this position**
- IMPORTANT: There must be NO air between probe and tank**



Standard for best installation position

- Probe Show stable **GREEN** Light (if flash red, you need hold and recheck other position)
- Mobile APP meets the following status:





- Real Time & Smooth Value are stable & close to real height
- Signal Strength more than 30 or stable (maximum 99)
- Valid Signal Quantity is stable, more than 30 (maximum 40)
- Tilt Angle is less than 11 Degrees (best 1~5 degrees)

5. Fix the probe to the bottom of tank for Metal/Steel Tank

- 1) Prepare AB Epoxy Glue (Suggested Ergo 9988, or other brands for metal with 3~5 minutes Curing Time)



OR



- 2) Inject the AB Epoxy Glue on the probe, and mix AB glue for 20~30 seconds to make sure glue color has changed.



- 3) Hold the probe with hand or Manual Jack/Lift for 10~15 minutes, LED must show Stable GREEN



4) Verify the installation

After step 3), remove your hand or Manual Jack/Lift, and use Mobile APP to check again to be sure the same as to complete the installation

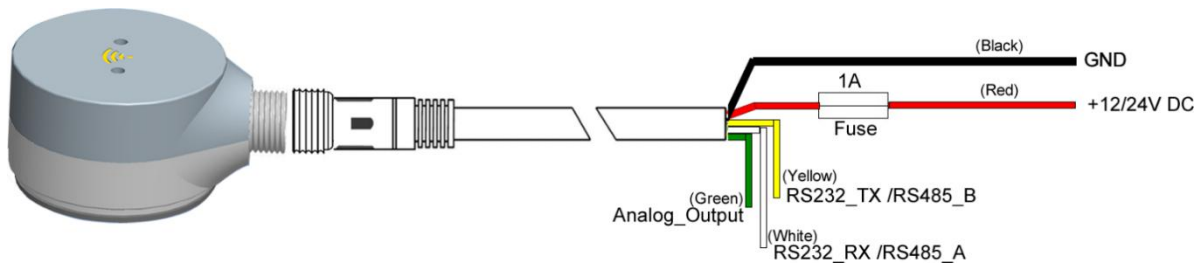
6. Wiring Diagram

Power Input: 9~36V DC



It is Mandatory to use an external Automotive 1 A Fuse

Recommended



Optional for LCV Vehicles

